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H5AG36EXNDX017N

H5AG36EXNDX017N

Industrial / Embedded Memory Solution

Technical Datasheet

Source Memory Chips (SMC) provides technical reference documentation for memory products used in industrial, networking, embedded and long-life electronic applications.

https://www.sourcememorychips.com

1. Product Overview

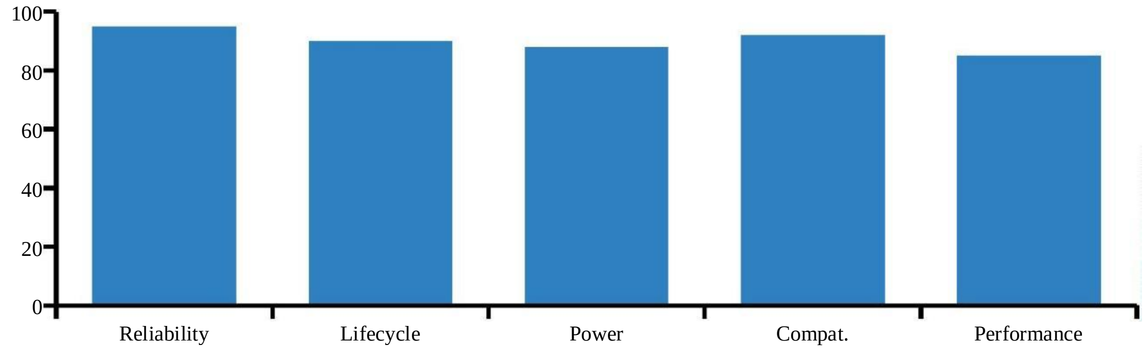
H5AG36EXNDX017N is an 8Gb DDR4 SDRAM memory device manufactured by SK hynix Electronics. The device is organized as 512M x16 bits and is intended for industrial, embedded, networking, and communication systems requiring stable memory performance and long product lifecycle support.

H5AG36EXNDX017N operates from a nominal 1.2V supply and uses DDR4 SDRAM architecture to provide a balanced combination of memory density, power efficiency, and controller compatibility. The x16 organization offers system designers a practical interface width for compact embedded boards and industrial computing platforms.

Parameter	Specification
Manufacturer	SK hynix Electronics
Part Number	H5AG36EXNDX017N
Memory Type	DDR4 SDRAM
Density	8Gb
Organization	512M x16
Supply Voltage	1.2V
Application Focus	Industrial / Embedded

Figure 1. H5AG36EXNDX017N feature profile

Representative feature emphasis for industrial embedded use



Key Takeaways

- SK hynix 8Gb DDR4 SDRAM device.
- 512M x16 organization for embedded memory subsystems.
- 1.2V operation for low-power DDR4 platforms.
- Positioned for industrial, networking, and long-life applications.

2. Ordering Information

The complete ordering code of H5AG36EXNDX017N should be reviewed during design qualification and procurement. Each segment of the code identifies product family, memory technology, density, organization, revision, and ordering suffix.

For H5AG36EXNDX017N, the code indicates a SK hynix DRAM product in the DDR4 SDRAM family, with 8Gb density and x16 organization. Procurement teams should verify the full code rather than relying only on the base model number.

H5AG36EXNDX017N part number structure

K	4	A	8G	16	WG	BCWE
SK hynix	DRAM	DDR4	8Gb	x16	Revision	Speed / code

Segment	Description
K	SK hynix memory product
4	DRAM product family
A	DDR4 SDRAM technology
8G	8Gb density
16	x16 data organization
WG	Device revision identifier
BCWE	Speed grade / ordering suffix code

Design Note

Before approving H5AG36EXNDX017N for production use, verify the complete ordering code, manufacturer marking, date code, traceability record, and the latest available manufacturer documentation.

Key Takeaways

- H5AG36EXNDX017N should be qualified using the complete ordering code.
- Suffix codes may affect speed grade, package, or product revision.
- Traceability and documentation review are important for long-life programs.

3. Device Overview

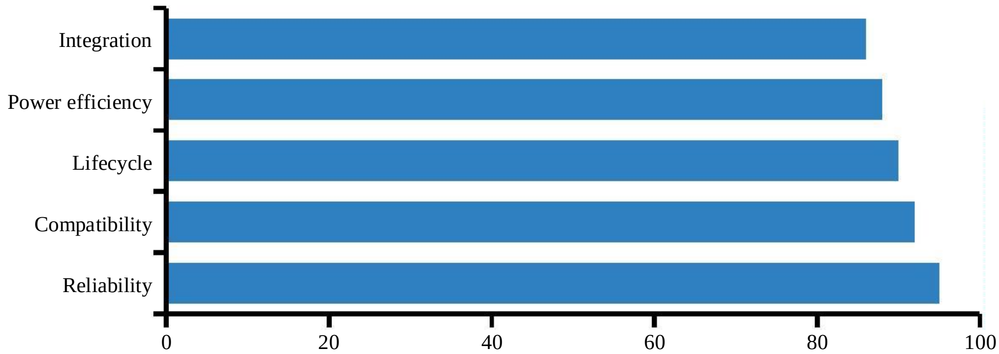
H5AG36EXNDX017N provides an 8Gb DDR4 SDRAM solution for systems that require stable memory capacity, power-efficient operation, and compatibility with established DDR4 memory controllers. The device is especially relevant for industrial computers, communication gateways, embedded controllers, and network infrastructure platforms.

The 512M x16 organization gives H5AG36EXNDX017N a practical balance between memory capacity and routing complexity. This makes the device suitable for compact PCB layouts where a x16 memory interface can reduce component count and simplify memory subsystem implementation.

Device Attribute	H5AG36EXNDX017N Value
Density	8Gb
Organization	512M x16
Technology	DDR4 SDRAM
Nominal Voltage	1.2V
Interface Width	x16
Target Systems	Industrial / Embedded / Networking

Figure 2. Device capability overview

Relative positioning for embedded design requirements



Key Takeaways

- H5AG36EXNDX017N is positioned as an 8Gb x16 DDR4 device.
- The device supports compact embedded and industrial designs.
- The main value is stable memory performance for long-life platforms.

4. Key Features

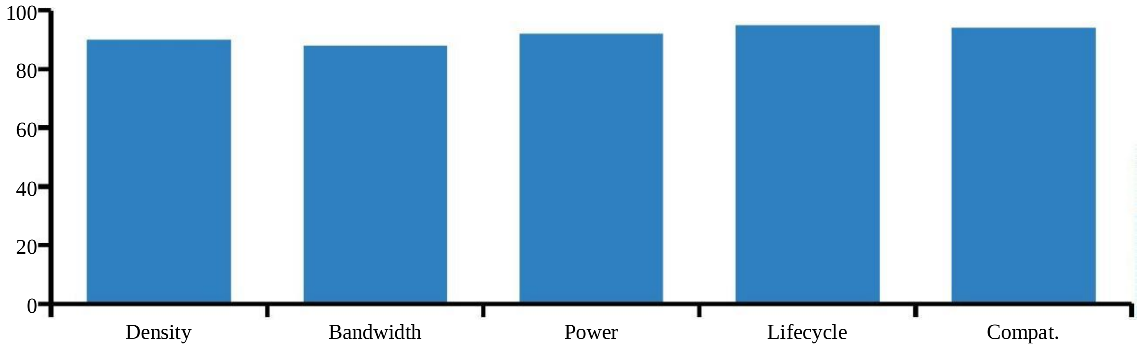
H5AG36EXNDX017N integrates a high-density DDR4 memory array with low-voltage operation and x16 organization. These characteristics make the device suitable for industrial and embedded systems requiring stable performance, efficient power usage, and broad DDR4 ecosystem compatibility.

The device is appropriate for platforms where lifecycle continuity, system reliability, and procurement traceability are key design and sourcing requirements. For engineers, H5AG36EXNDX017N offers a mature memory solution for applications that need proven DDR4 technology rather than experimental platform migration.

Feature	Description	System Benefit
8Gb density	High-capacity memory array	Supports larger workloads
x16 organization	16-bit data interface	Balanced routing and performance
1.2V operation	Low-voltage DDR4 supply	Reduced system power
DDR4 architecture	Mature memory ecosystem	Lower integration risk
Industrial use	Long-life platform suitability	Improved lifecycle planning

Figure 3. Key feature emphasis

Representative strengths of H5AG36EXNDX017N



Feature Summary

- High-density 8Gb DDR4 SDRAM device.
- x16 organization supports practical embedded memory designs.
- 1.2V operation supports lower-power system architectures.
- Suitable for industrial, networking, embedded, and long-life applications.

5. Architecture Overview

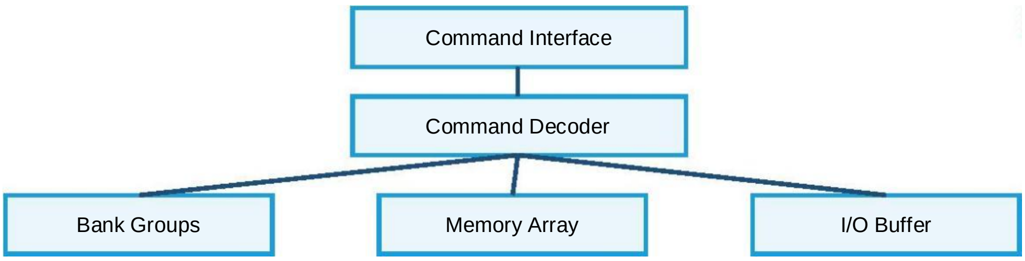
H5AG36EXNDX017N is based on SK hynix DDR4 SDRAM architecture and is organized as a high-density 8Gb memory device with a x16 data interface. The device architecture is designed to balance memory capacity, performance, and power efficiency for industrial and embedded computing platforms requiring stable long-term operation.

The internal structure of H5AG36EXNDX017N supports efficient memory access and bandwidth utilization while maintaining compatibility with widely adopted DDR4 memory controllers. This enables system designers to implement reliable memory subsystems without introducing unnecessary integration complexity.

Parameter	Specification
Part Number	H5AG36EXNDX017N
Density	8Gb
Organization	512M x16
Interface Width	x16
Operating Voltage	1.2V
Application Focus	Industrial / Embedded

Figure 4. Architecture overview

H5AG36EXNDX017N simplified architecture



Architecture Feature	System Benefit
8Gb density	Supports larger memory workloads
x16 organization	Simplifies embedded memory routing
DDR4 architecture	Provides mature ecosystem compatibility
Low-voltage operation	Improves system power efficiency

Key Takeaways

- H5AG36EXNDX017N uses a mature DDR4 SDRAM architecture.
- The 512M x16 organization supports compact embedded designs.
- The device is suitable for industrial systems requiring stable long-term memory operation.

6. Electrical Characteristics

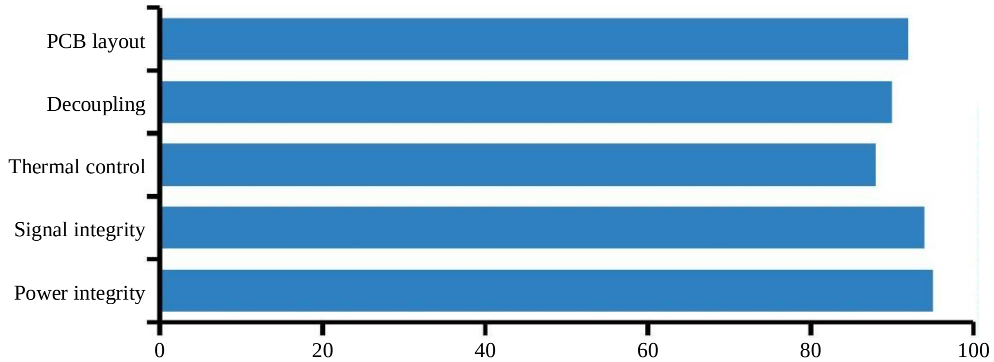
H5AG36EXNDX017N operates from a nominal 1.2V power supply and is optimized for low-power DDR4 memory applications. Stable power delivery and signal integrity are important design considerations to ensure reliable device operation in industrial and embedded platforms.

The low-voltage DDR4 architecture helps reduce overall system power consumption while maintaining the bandwidth required by embedded processors and communication platforms. Proper decoupling, controlled impedance routing, and clean reference planes help the device maintain predictable memory behavior.

Parameter	Symbol	Typical
Core voltage	VDD	1.2V
I/O voltage	VDDQ	1.2V
Memory density	-	8Gb
Organization	-	512M x16
Technology	-	DDR4 SDRAM

Figure 5. Electrical design emphasis

Relative importance for stable H5AG36EXNDX017N operation



Design Area	Recommendation
Power delivery	Maintain stable 1.2V supply rails
Signal routing	Use controlled impedance practices
Grounding	Provide continuous return paths
Validation	Test under worst-case system loading

Key Takeaways

- 1.2V low-voltage DDR4 operation.
- Power and signal integrity are central to reliable deployment.
- H5AG36EXNDX017N should be validated at the system level.

7. Operating Conditions

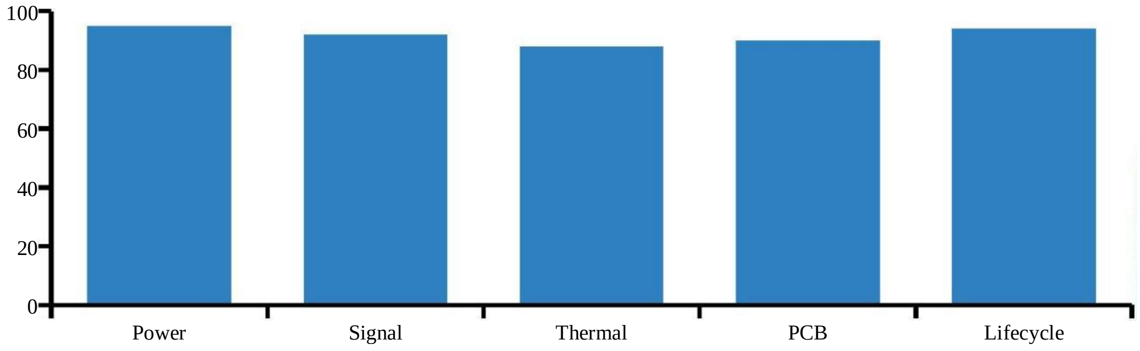
H5AG36EXNDX017N is intended for deployment in industrial and embedded systems where reliable memory operation is required throughout the product lifecycle. System designers should maintain recommended electrical, thermal, and layout conditions to achieve stable operation.

In addition to electrical design, environmental conditions such as airflow planning, operating temperature control, and PCB assembly quality contribute to long-term reliability. H5AG36EXNDX017N is commonly selected for systems that operate continuously and require predictable memory behavior.

Category	Description
Deployment type	Continuous operation
Application class	Industrial / Embedded
Lifecycle focus	Long-term support
Reliability priority	High
Design objective	Stable memory performance

Figure 6. Operating stability profile

H5AG36EXNDX017N operating condition priorities



Requirement	Importance
Power stability	High
Signal integrity	High
Thermal management	High
PCB design quality	High
Lifecycle planning	High

Application Highlights

- Suitable for industrial computers and embedded controllers.
- Applicable to networking equipment and communication gateways.
- Designed for systems requiring stable long-term memory operation.

8. Industrial & Embedded Applications

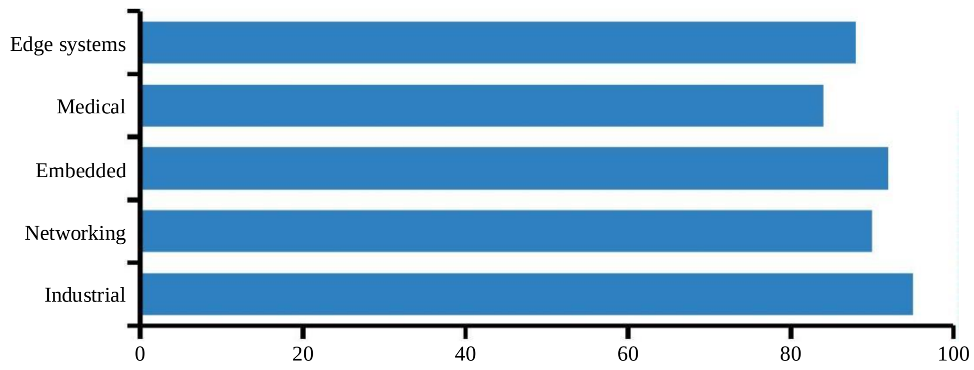
H5AG36EXNDX017N is widely used in industrial and embedded computing platforms where memory reliability, supply continuity, and stable long-term performance are important requirements. Its 8Gb density and x16 organization support a broad range of memory subsystem configurations.

The device is suitable for industrial automation, networking infrastructure, embedded computing, medical electronics, security systems, and edge devices. These applications typically value mature memory technology, dependable sourcing, and compatibility with established DDR4 platforms.

Application Area	Typical Systems
Industrial automation	PLC, HMI, motion controllers
Networking equipment	Switches, routers, firewalls
Embedded computing	SBCs, industrial PCs
Medical electronics	Diagnostic and monitoring systems
Edge computing	Industrial gateways

Figure 7. Typical application fit

Representative suitability by application segment



Deployment Characteristic	Priority
Lifecycle support	High
Reliability	High
Power efficiency	High
DDR4 compatibility	High
Supply stability	High

Key Takeaways

- H5AG36EXNDX017N fits industrial and embedded platforms.
- The device supports long-life DDR4 system designs.
- Applications should verify compatibility, lifecycle status, and supply traceability.

9. Package Information

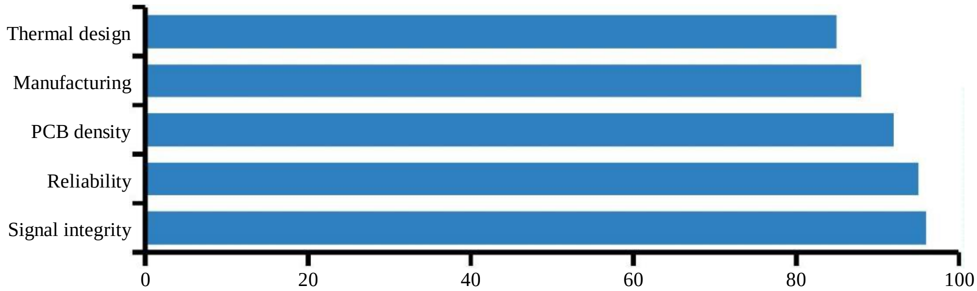
H5AG36EXNDX017N is supplied in a compact FBGA package optimized for industrial and embedded DDR4 memory applications. The package architecture supports high-density PCB integration while maintaining signal integrity, manufacturability, and long-term reliability.

The FBGA structure minimizes signal path length and supports efficient routing, making the device suitable for networking equipment, industrial computers, embedded controllers, and communication gateways.

Parameter	Specification
Part Number	H5AG36EXNDX017N
Package Type	FBGA
Memory Type	DDR4 SDRAM
Density	8Gb
Organization	512M x16
Application	Industrial / Embedded

Figure 8. Package Deployment Priorities

Representative package integration priorities



Design Area	Recommendation
Routing	Controlled impedance traces
Grounding	Continuous reference planes
Power Integrity	Stable VDD/VDDQ rails
Decoupling	Local bypass capacitors
Thermal Design	Adequate airflow planning

Key Takeaways

- Compact FBGA package for embedded DDR4 systems.
- Optimized for signal integrity and dense PCB layouts.
- Suitable for industrial, networking, and long-life deployment.

10. Quality & Reliability

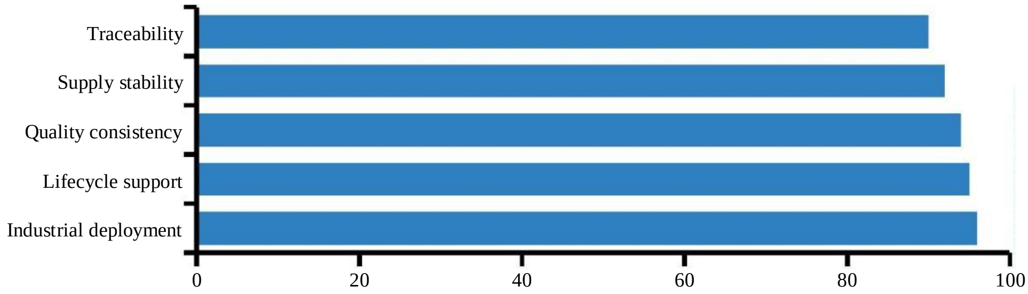
H5AG36EXNDX017N is commonly deployed in industrial and embedded systems requiring stable long-term memory operation. Reliability, lifecycle support, and supply continuity are important considerations throughout product deployment.

System designers should validate compatibility, thermal behavior, and power delivery during platform qualification to ensure predictable field performance in continuous-operation environments.

Category	Objective
Qualification	Platform verification
Validation	Stable operation
Traceability	Supply visibility
Lifecycle Planning	Long-term support
Procurement	Risk mitigation

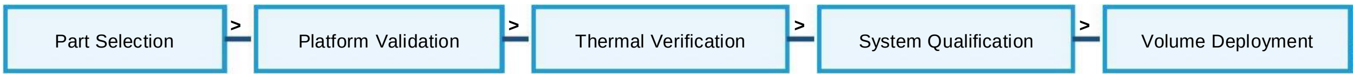
Figure 9. Reliability Focus Profile

Representative priorities for long-life industrial deployment



Verification Item	Status
Part Number Review	Required
Supplier Verification	Required
PCB Validation	Required
Thermal Testing	Required
Lifecycle Assessment	Recommended

Qualification Flow



Key Takeaways

- Industrial deployment focus with strong lifecycle planning.
- Traceability and validation are critical for production approval.
- Suitable for long-life embedded and networking programs.

11. Conclusion

H5AG36EXNDX017N is an 8Gb DDR4 SDRAM memory device organized as 512M x16 and optimized for industrial, networking, communication, and embedded applications.

Its mature DDR4 architecture, low-voltage operation, and broad ecosystem compatibility make it a practical solution for long-life deployments that require reliable memory performance and stable sourcing.

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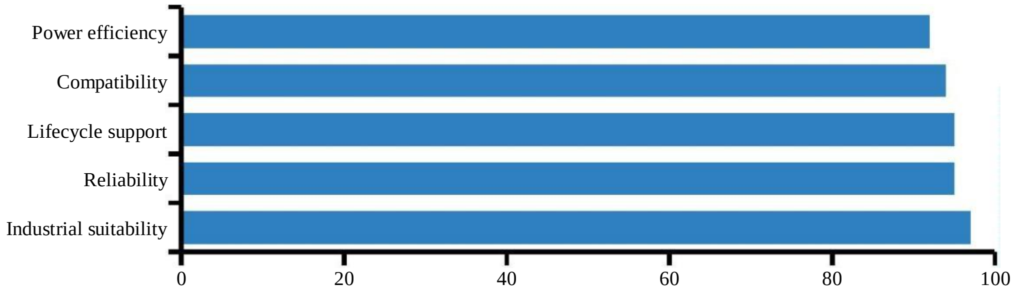
Interface Width	x16
Application Focus	Industrial / Embedded

Figure 10. Overall Device Profile

Representative strengths across key deployment criteria



Segment	Example System
Industrial Automation	PLC / HMI
Networking Equipment	Router / Switch
Embedded Computing	Industrial PC
Medical Electronics	Diagnostic Systems
Edge Computing	Industrial Gateway

Final Key Takeaways

- SK hynix 8Gb DDR4 SDRAM solution with 512M x16 organization.
- 1.2V low-voltage operation for mature DDR4 platforms.
- Focused on industrial, embedded, and long-life deployment needs.